in his wonderful book *Creation Revisited*, uses a similar personification when considering the refraction of a light beam, passing into a medium of higher refractive index which slows it down. The beam behaves as if trying to minimize the time taken to travel to an end point. Atkins imagines it as a lifeguard on a beach racing to rescue a drowning swimmer. Should he head straight for the swimmer? No, because he can run faster than he can swim and would be wise to increase the dry-land proportion of his travel time. Should he run to a point on the beach directly opposite his target, thereby minimizing his swimming time? Better, but still not the best. Calculation (if he had time to do it) would disclose to the lifeguard an optimum intermediate angle, yielding the ideal combination of fast running followed by inevitably slower swimming. Atkins concludes:

> That is exactly the behaviour of light passing into a denser medium. But how does light know, apparently in advance, which is the briefest path? And, anyway, why should it care?

He develops these questions in a fascinating exposition, inspired by quantum theory.

Personification of this kind is not just a quaint didactic device. It can also help a professional scientist to get the right answer, in the face of tricky temptations to error. Such is the case with Darwinian calculations of altruism and selfishness, cooperation and spite. It is very easy to get the wrong answer. Personifying genes, if done with due care and caution, often turns out to be the shortest route to rescuing a Darwinian theorist drowning in muddle. While trying to exercise that caution, I was encouraged by the masterful precedent of W. D. Hamilton, one of the four named heroes of the book. In a paper of 1972 (the year in which I began to write *The Selfish Gene*) Hamilton wrote:

> A gene is being favoured in natural selection if the aggregate of its replicas forms an increasing fraction of the total gene pool. We are going to be concerned with genes supposed to affect the social behaviour of their bearers, so let us try to make the argument more vivid by attributing to the genes, temporarily, intelligence and a certain freedom of choice. Imagine that a gene is considering the problem of increasing the number of its replicas, and imagine that it can choose between …

That is exactly the right spirit in which to read much of *The Selfish Gene*.

Personifying an organism could be more problematical. This is because organisms, unlike genes, have brains and therefore really might